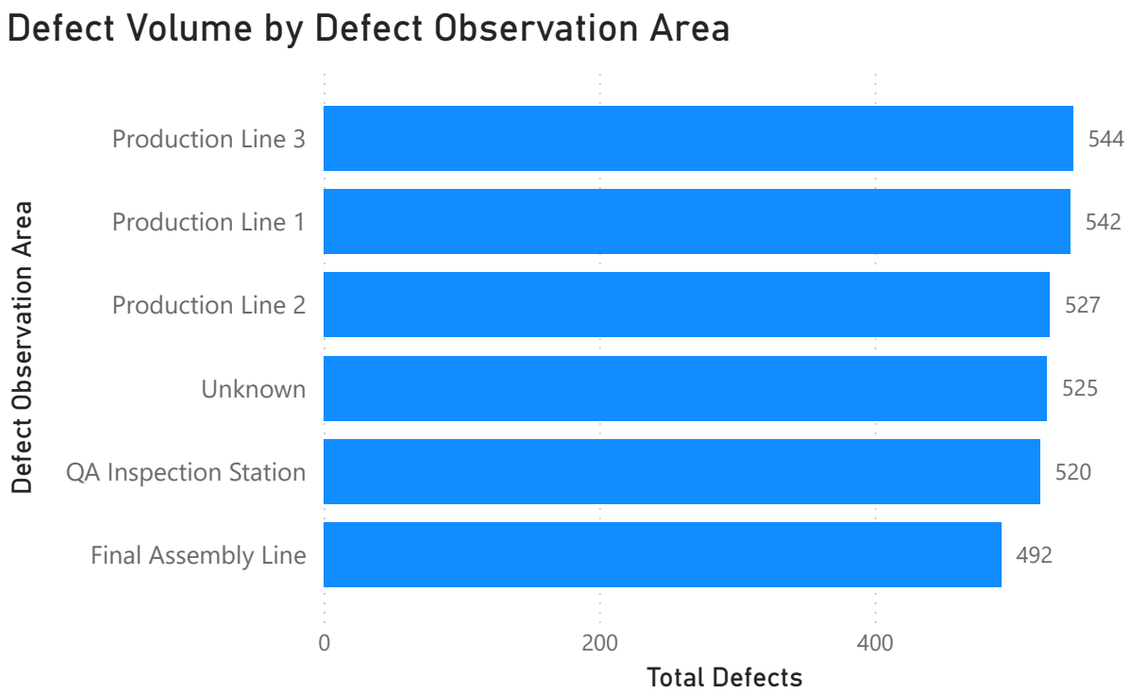
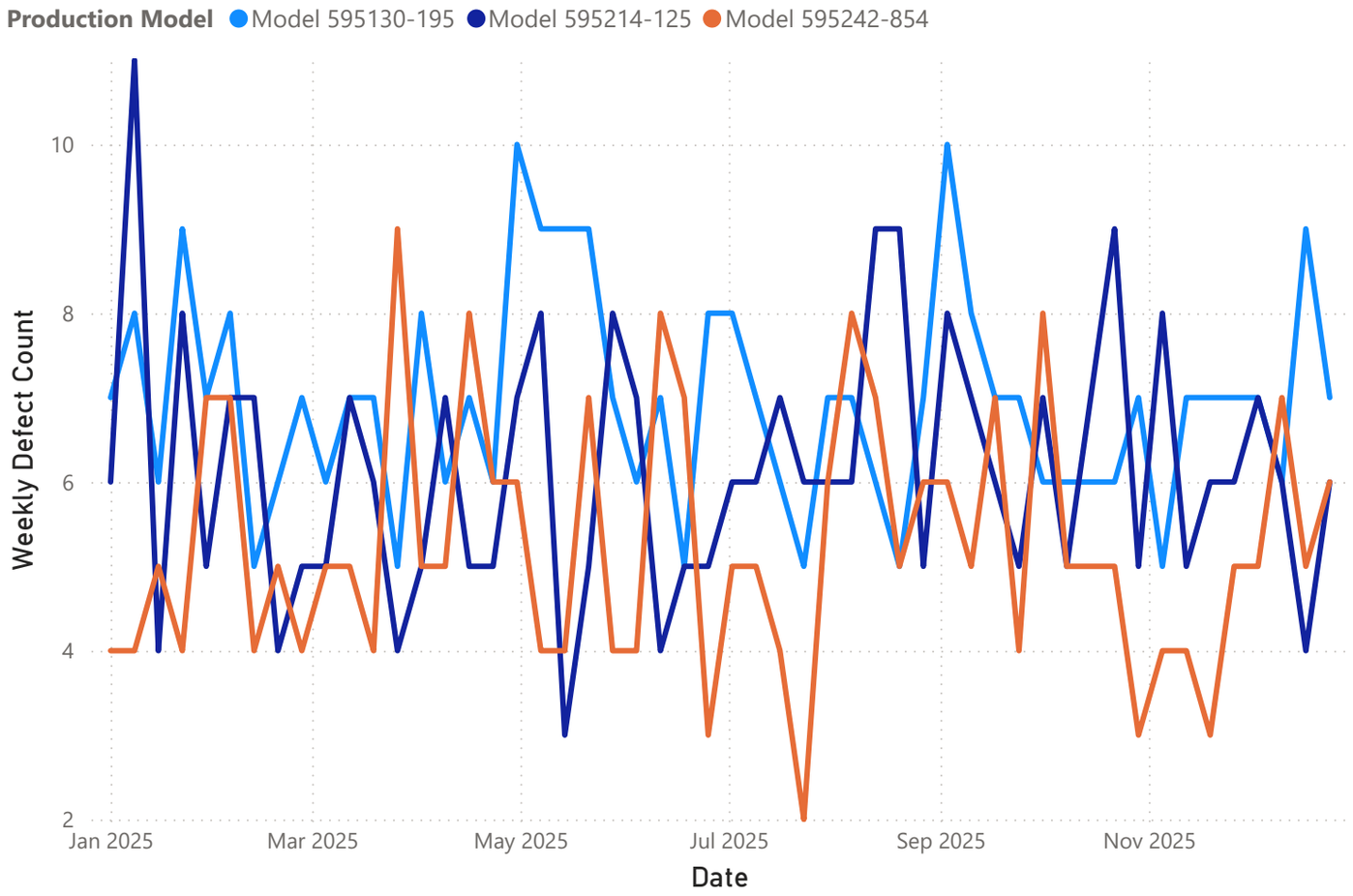

Defect Volume by Defect Observation Area
Defect Observation Area
Production Line 3
Production Line 1
Production Line 2
Unknown
QA Inspection Station
Final Assembly Line
544
542
527
525
520
492
0
200
400
Total Defects

Weekly Defect Trend - Top 3 Production Models



F-Statistic (ANOVA)

17.64

F_Statistic

p-value (ANOVA)

p< 0.001

p < 0.001

Statistical Significance

F-Statistic (ANOVA)

17.64

F_Statistic

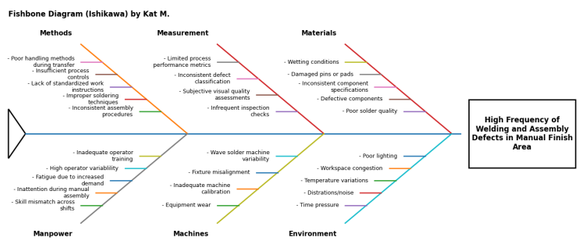
p-value (ANOVA)

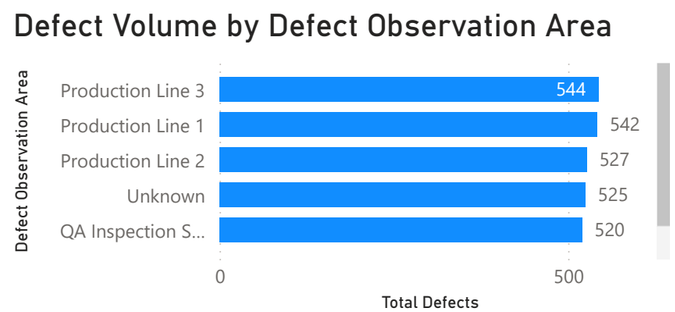
p< 0.001

$p < 0.001$

Statistical Significance

Manufacturing Defect Analysis Dashboard





F-Statistic (ANOVA)

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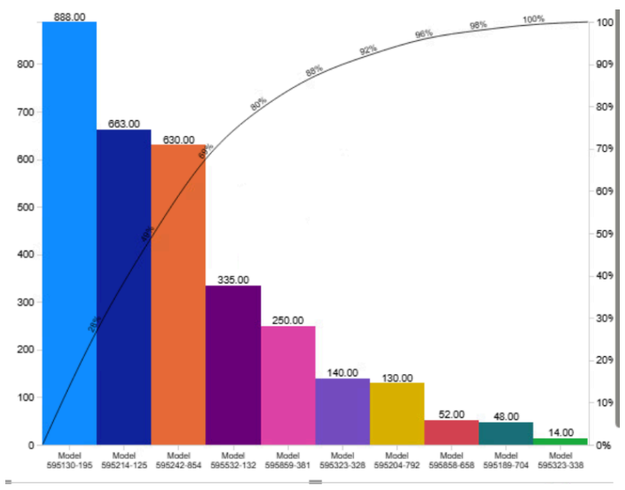
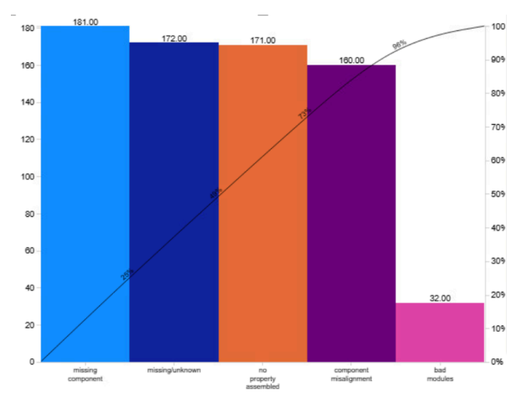
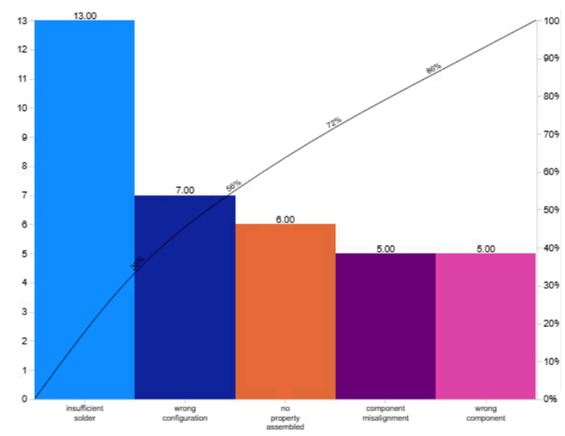
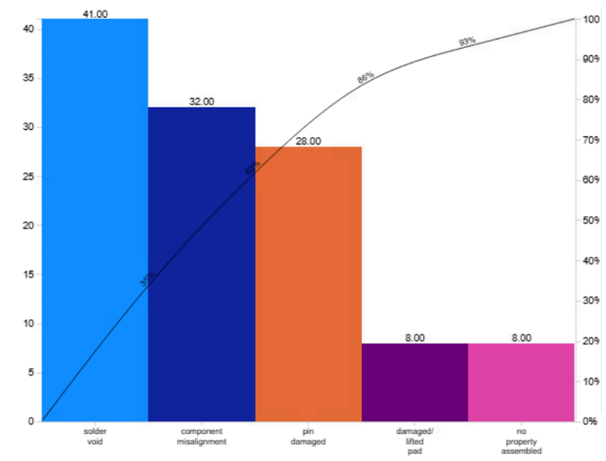
Statistical Significance

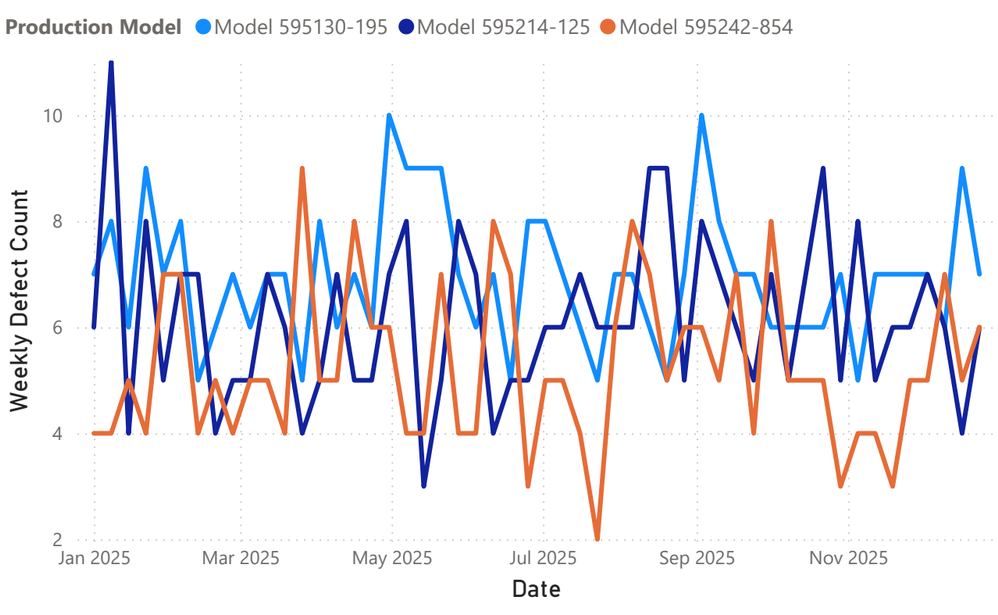








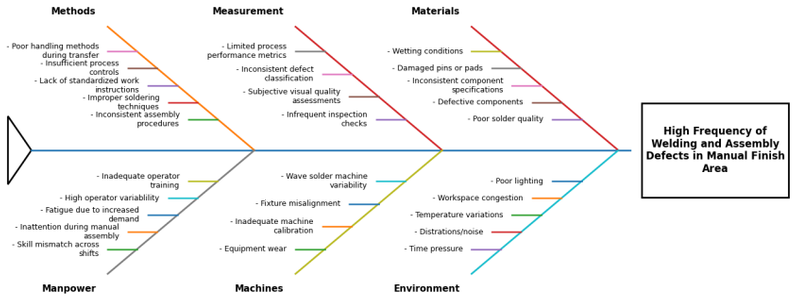
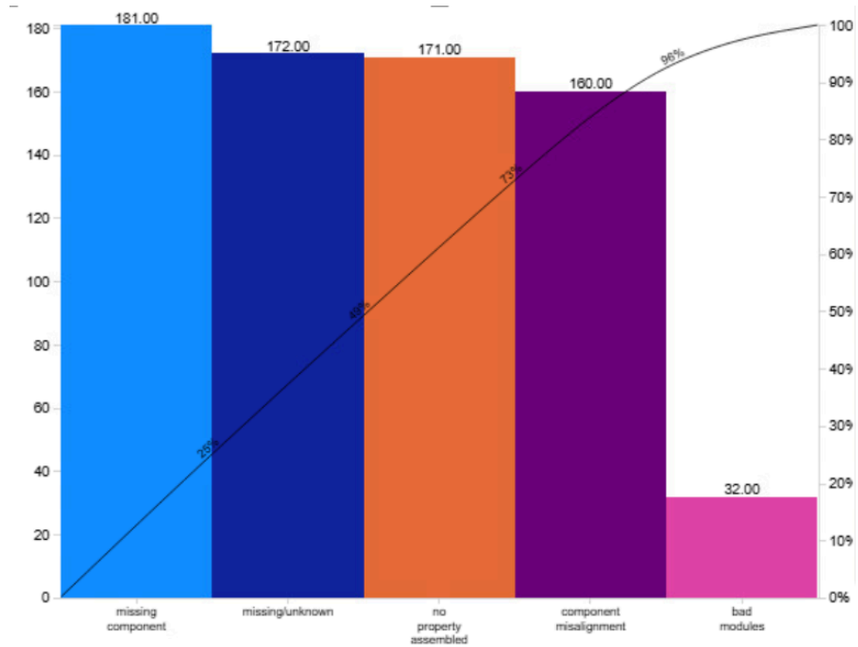
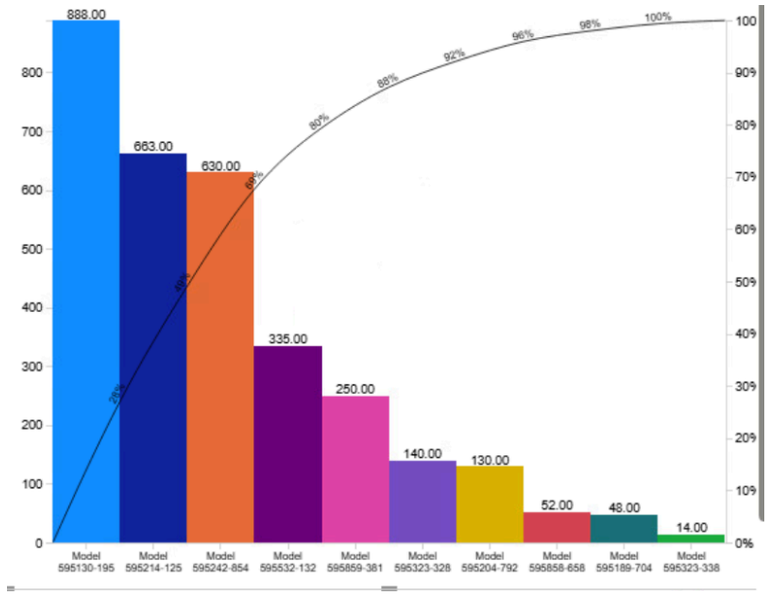
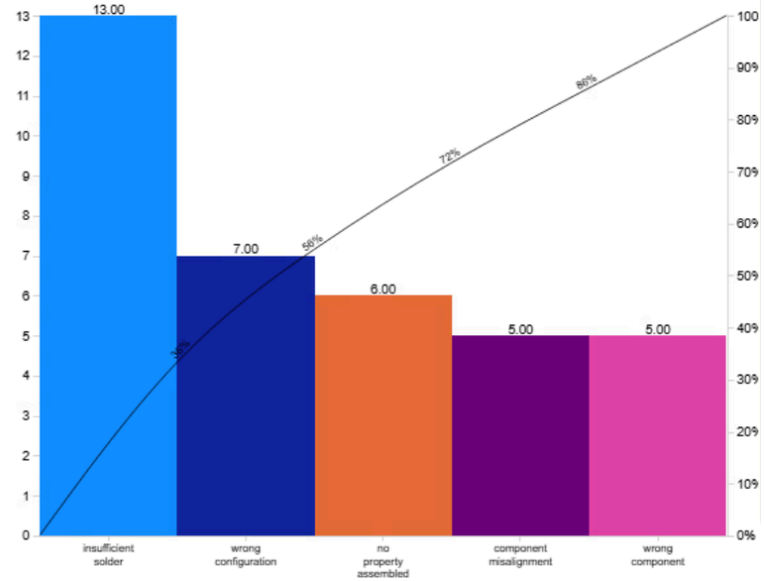
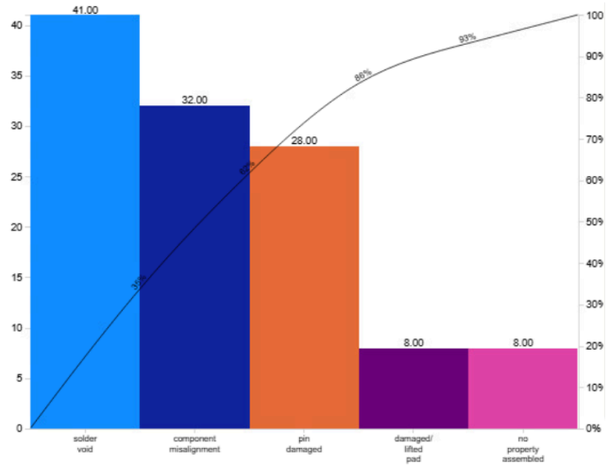
Weekly Defect Trend - Top 3 Production Models



Conclusion

Root cause analysis identified concentration of defects across key production areas, with Production Line 3 contributing the highest defect volume. Weekly trend analysis showed consistent fluctuation across the top three production models, with Model 595130-195 demonstrating the highest average defect count.

A one-way ANOVA test confirmed statistically significant differences in mean weekly defect counts among the three models ($F = 17.64$, $p < 0.001$). The null hypothesis was rejected, indicating that production model type significantly influences defect variation.

Based on these finding, process improvement efforts should prioritize targeted review and quality control interventions for Model 595130-195, along with continued monitoring of high-volume production areas to reduce overall defect rates.

Fishbone Diagram (Ishikawa) by Kat M.